The empirical recurrence for OEIS A302152, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A302152 records "Empirical recurrence of order 85" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 118$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 90$. Its last coefficient is $c_{85} = -16 \neq 0$, so no further tail satisfies anything shorter; any order-85 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A302152 is "Number of 4 X n 0..1 arrays with every element equal to 0, 2, 3 or 4 horizontally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$8, 1, 10, 17, 56, 255, 1038, 4771, \ \ldots$$

The entry states:

> Empirical recurrence of order 85 (see link above).

As of the "Last modified" line on the live entry (Aug 19 2022 15:35:41, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \; = \; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 118$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 118$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 85: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 85.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 236$ exact terms from $n = 5$ on, it returns order 85 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{85} c_i\, a(n-i),$$

c_1	8	c_{23}	−678101	c_{45}	−51638593	c_{67}	−1217521
c_2	−20	c_{24}	−1138907	c_{46}	−53889138	c_{68}	−636282
c_3	28	c_{25}	−1863224	c_{47}	−48428279	c_{69}	−112459
c_4	−35	c_{26}	−2668165	c_{48}	−37402720	c_{70}	−11129
c_5	30	c_{27}	−3420598	c_{49}	−22282785	c_{71}	−75278
c_6	−56	c_{28}	−4077271	c_{50}	−6070041	c_{72}	−29795
c_7	−53	c_{29}	−4108834	c_{51}	8438087	c_{73}	59726
c_8	−164	c_{30}	−3385989	c_{52}	18800403	c_{74}	86140
c_9	−529	c_{31}	−1441917	c_{53}	23798261	c_{75}	65854
c_{10}	−472	c_{32}	2327701	c_{54}	24669671	c_{76}	40868
c_{11}	624	c_{33}	7349542	c_{55}	22744647	c_{77}	22507
c_{12}	1375	c_{34}	13730142	c_{56}	18466932	c_{78}	10840
c_{13}	6546	c_{35}	20827249	c_{57}	12517718	c_{79}	4626
c_{14}	15911	c_{36}	27218844	c_{58}	7347481	c_{80}	1232
c_{15}	26068	c_{37}	31745206	c_{59}	3853257	c_{81}	−580
c_{16}	50155	c_{38}	32173704	c_{60}	1665642	c_{82}	−908
c_{17}	66200	c_{39}	28592987	c_{61}	262032	c_{83}	−492
c_{18}	78681	c_{40}	20065686	c_{62}	−1103012	c_{84}	−136
c_{19}	87767	c_{41}	6189363	c_{63}	−2330396	c_{85}	−16
c_{20}	23302	c_{42}	−10104160	c_{64}	−2445303		
c_{21}	−74008	c_{43}	−27167567	c_{65}	−1837583		
c_{22}	−291778	c_{44}	−42028134	c_{66}	−1470614		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 90$, and it is the recurrence OEIS A302152 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{85} - c_1 x^{85-1} - \cdots - c_{85}$. Its constant term is $-c_{85} = 16$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 85 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 85, so $\deg q = 85$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 85, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 85, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -16.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-85 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A302152.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.